



DSA MARATHON

Part 1 — Questions 1–10

With Complete C++ Programs

1. Complete Student Report

Take Name, Roll Number, and marks of 5 subjects. Print Total, Percentage, Average in a formatted report.

ANSWER

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string name;
    int roll;
    int m1, m2, m3, m4, m5;
    getline(cin, name);
    cin >> roll;
    cin >> m1 >> m2 >> m3 >> m4 >> m5;
    int total = m1 + m2 + m3 + m4 + m5;
    double percentage = total / 5.0;
    double average = total / 5.0;
    cout << "===== STUDENT REPORT =====" << endl;
    cout << "Name : " << name << endl;
    cout << "Roll No : " << roll << endl;
    cout << "Total : " << total << endl;
    cout << "Percentage : " << percentage << "%" << endl;
    cout << "Average : " << average << endl;
    return 0;
}
```

2. Salary Calculator

Employee Name, Basic Salary. HRA=20%, DA=10%, Gross Salary.

ANSWER

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string name;
    double basic;
    getline(cin, name);
    cin >> basic;
    double hra = basic * 0.20;
    double da = basic * 0.10;
    double gross = basic + hra + da;
    cout << "Gross Salary = " << gross << endl;
}
```

```
    return 0;
}
```

3. Electricity Bill Generator

*Consumer Name, Units. Bill=Units*7.5, GST 18%, Final.*

ANSWER

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string name;
    double units;
    getline(cin,name);
    cin>>units;
    double bill=units*7.5;
    double gst=bill*0.18;
    double finalBill=bill+gst;
    cout<<"Bill = "<<bill<<endl;
    cout<<"GST = "<<gst<<endl;
    cout<<"Final = "<<finalBill<<endl;
    return 0;
}
```

4. Restaurant Bill Generator

Customer name and prices of 4 items.

ANSWER

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string name;
    double a,b,c,d;
    getline(cin,name);
    cin>>a>>b>>c>>d;
    double total=a+b+c+d;
    double gst=total*0.05;
    cout<<"Total = "<<total<<endl;
    cout<<"GST = "<<gst<<endl;
    cout<<"Final = "<<total+gst<<endl;
    return 0;
}
```

5. Bank Account Summary

Account holder, account no, balance, deposit.

ANSWER

```
#include <iostream>
#include <string>
```

```
using namespace std;
int main()
{
    string name, account;
    double balance, deposit;
    getline(cin,name);
    getline(cin,account);
    cin>>balance>>deposit;
    cout<<"Updated Balance = "<<balance+deposit<<endl;
    return 0;
}
```

6. Product Billing System

Product, quantity, price, total, 10% discount.

ANSWER

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string product;
    int quantity;
    double price;
    getline(cin,product);
    cin>>quantity>>price;
    double total=quantity*price;
    double finalPrice=total-(total*0.10);
    cout<<"Total = "<<total<<endl;
    cout<<"Final = "<<finalPrice<<endl;
    return 0;
}
```

7. Travel Expense Calculator

Distance, efficiency, petrol price.

ANSWER

```
#include <iostream>
using namespace std;
int main()
{
    double distance, efficiency, petrolPrice;
    cin>>distance>>efficiency>>petrolPrice;
    double fuel=distance/efficiency;
    double cost=fuel*petrolPrice;
    cout<<"Fuel = "<<fuel<<endl;
    cout<<"Cost = "<<cost<<endl;
    return 0;
}
```

8. Time Converter

Convert minutes to hours and minutes.

ANSWER

```
#include <iostream>
using namespace std;
int main()
{
    int minutes;
    cin>>minutes;
    cout<<minutes/60<<" Hours "<<minutes%60<<" Minutes"<<endl;
    return 0;
}
```

9. Currency Converter

INR to USD, EUR, GBP.

ANSWER

```
#include <iostream>
using namespace std;
int main()
{
    double inr;
    cin>>inr;
    cout<<"USD = "<<inr/86.0<<endl;
    cout<<"EUR = "<<inr/100.0<<endl;
    cout<<"GBP = "<<inr/116.0<<endl;
    return 0;
}
```

10. Character Analyzer

Character, ASCII, next, previous.

ANSWER

```
#include <iostream>
using namespace std;
int main()
{
    char ch;
    cin>>ch;
    cout<<"Character = "<<ch<<endl;
    cout<<"ASCII = "<<int(ch)<<endl;
    cout<<"Next = "<<char(ch+1)<<endl;
    cout<<"Previous = "<<char(ch-1)<<endl;
    return 0;
}
```

End of Part 1 Question Set

NextGen Data Minds